AI Automation & Agents

Customer email triage and replies · Demo Client

Prepared for	Demo Client
Engagement	DEMO-0013
Report date	13 November 2026
Version	1.0
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Run ID	aa_20261113T113019Z_9ce23d

Executive summary

Demonstration report. Built from synthetic data generated by ProQuant for illustration: an online retailer that does not exist. The agent's answers come from a simulated model, not from a call to a real AI service. No real business, customer or person is described.

ProQuant Analytics looked at every process Demo Client's teams run by hand to find where automation pays, then built an AI agent for customer email triage and replies and measured it on 400 messages the team had labelled beforehand, followed by an 8-week pilot beside the team. The agent reads a message, names the request and drafts a reply; a fixed policy, not the model, decides whether a person must see it first.

Go-live verdict: Ready, with a person reviewing

11 of 15 go-live tests passed, 4 with a warning, 0 failed. Thresholds and the confidence bar were set before any result was seen.

INTENT RIGHT	ANSWERED WITHOUT A PERSON	HOURS GIVEN BACK A MONTH
92.2%	**26%**	**154**
400 labelled cases, 90% to 95%	at confidence 0.90, 0 unsafe	customer email triage and replies

AI COST PER MESSAGE	PILOT DRAFTS CHANGED	PERSONAL DETAILS MASKED
USD 0.0088	**20%**	**510**
against USD 2.12 by hand	2,411 messages in 8 weeks	0 left after masking

Findings

- **Where** Of 14 processes run by hand, 4 are quick wins worth USD 144k a year together; Contract clause review and Credit limit approvals for B2B customers should stay with a person.
- **Quality** The agent named the intent right in 92.2% of 400 labelled messages; rules alone managed 59.0%.
- **Safety** At the agreed confidence of 0.90, 26% of messages are answered without a person and none of them should not have been.
- **Threshold** A lower bar of 0.80 would have answered 38% alone, including 3 that a person had to handle.
- **Complaints** 11% of complaints were read as another request; until this improves, complaint words send a message to a person.
- **Money** For customer email triage and replies, the agent gives back about 154 hours a month for USD 123 in running costs.
- **Pilot** Over 8 weeks, reviewers changed 20% of the drafts they saw, and 7 automatic replies needed a correction afterwards.

Actions

Ref	Priority	Action	Owner	Due
A-01	Medium	Read the edited drafts with the reviewers and tighten the reply rules in the prompt.	Customer service lead	2026-12-13
A-02	Medium	Check order numbers against the order system before any reply uses them.	IT	2026-12-13
A-03	Medium	Extend the keyword rules so the service still runs, more slowly, if the AI is switched off.	Customer service lead	2026-12-13
A-04	Medium	Send every message with complaint words to a person, whatever the model says, until recall passes.	Customer service lead	2026-12-13

Source: tables/actions.csv

Methodology

This report says what was found and what it means for the service. The methodology document explains how each measure is computed; the agent specification describes the instructions, the answer format and the policy. Read one idea before the results: a case counts as unsafe when the agent would have answered it alone although the team's own label says a person should handle it.

Results

1. Where automation pays

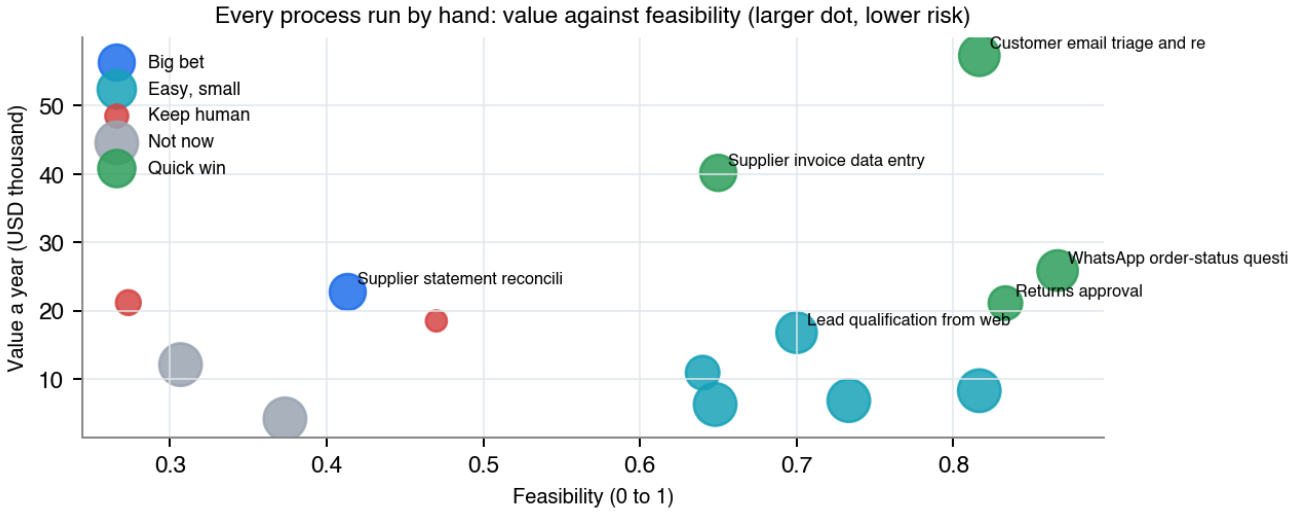


Figure 1. Each process by the value automation could free a year and how feasible it is; colour shows the verdict. Source: tables/opportunities.csv

#	Process	Team	Verdict	Hours a year by hand	Value a year (USD)	Build (USD)	Payback, months
1	Customer email triage and replies	Customer service	Quick win	3,120	57,283	5,400	1.16
2	Supplier invoice data entry	Finance	Quick win	1,960	40,141	5,400	1.67
3	WhatsApp order-status questions	Customer service	Quick win	1,330	25,855	5,400	2.65
4	Returns approval	Operations	Quick win	1,040	21,091	5,400	3.27
5	Supplier statement reconciliation	Finance	Big bet	1,080	22,810	13,500	7.50
6	Lead qualification from web forms	Sales	Easy, small	810	16,767	2,700	2.09
7	Staff HR questions	People	Easy, small	520	10,920	2,700	3.34
8	Stock reorder alerts	Operations	Easy, small	360	8,284	2,700	4.60
9	Product review replies	Marketing	Easy, small	450	6,831	2,700	5.87
10	Daily sales report	Finance	Easy, small	220	6,266	2,700	6.40
11	Contract clause review	Legal	Keep human	630	21,168	3,600	2.16
12	Credit limit approvals for B2B customers	Finance	Keep human	560	18,480	3,600	2.50
13	Meeting notes and follow-ups	Management	Not now	640	12,096	0	n/a
14	Social media posting	Marketing	Not now	300	4,158	0	n/a

Source: tables/opportunities.csv

Value counts only the share of the work automation can take, at the team's own hourly cost. Processes whose decisions carry high risk (credit, contracts) stay with a person even when the value is large: an AI may gather and summarise the file, but a person decides.

2. The agent, measured

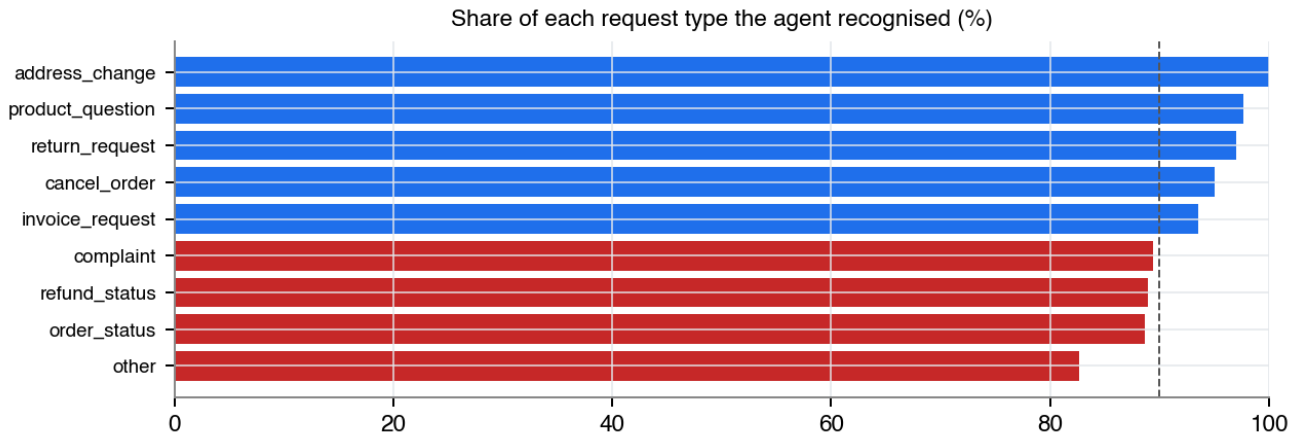


Figure 2. Recall by request type: the share of each type the agent recognised. Source: tables/per_intent.csv

Request type	Cases	Precision	Recall	F1
address_change	27	93.1%	100.0%	96.4%
cancel_order	20	100.0%	95.0%	97.4%
complaint	47	95.5%	89.4%	92.3%
invoice_request	31	90.6%	93.5%	92.1%
order_status	97	87.8%	88.7%	88.2%
other	23	90.5%	82.6%	86.4%
product_question	43	89.4%	97.7%	93.3%
refund_status	45	93.0%	88.9%	90.9%
return_request	67	97.0%	97.0%	97.0%

Source: tables/per_intent.csv

Side by side with the team and with keyword rules that use no AI:

Who answers	Intent right	Answered alone	Unsafe	Cost a message (USD)	Seconds
Team, by hand	n/a	0%	0	2.1200	318.00
Agent (simulated model)	92.2%	26%	0	0.0088	1.86
Rules, no AI	59.0%	0%	0	0.0000	0.00

Source: tables/engines.csv

The mistakes that matter most, by how often they occurred:

The request was	The agent read	Cases
refund_status	order_status	5
complaint	order_status	5
order_status	product_question	3
order_status	complaint	2
order_status	other	2
cancel_order	order_status	1
other	order_status	1
return_request	invoice_request	1

Source: tables/confusion.csv

Note: in this demonstration the agent's answers come from a simulated model built into the sample data. At a client, the same evaluation runs on the real model's recorded answers.

3. How much it may answer alone

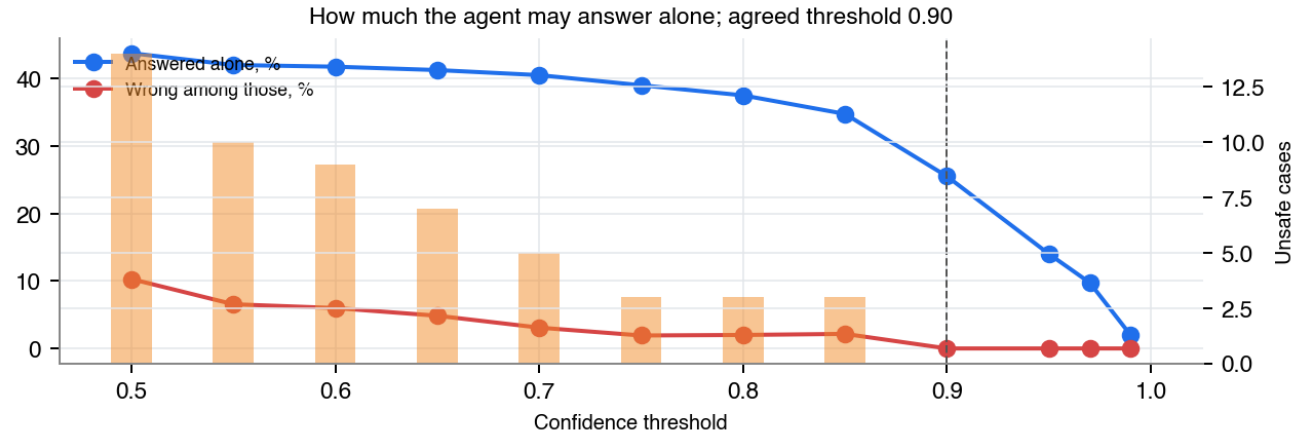


Figure 3. For each confidence threshold: the share answered without a person, the share of those that were wrong, and unsafe cases.
Source: tables/threshold_curve.csv

Threshold	Answered alone	Wrong among those	Unsafe	Sent to review
0.80	38%	2.0%	3	46%
0.90	26%	0.0%	0	58%
0.95	14%	0.0%	0	70%
0.99	2%	0.0%	0	82%

Source: tables/threshold_curve.csv

Raising the bar moves messages from automatic replies to review: safer, but fewer hours saved. At the agreed bar of 0.90 the test found 0 unsafe cases.

4. Privacy and the policy

Before any message reaches the model, e-mail addresses, phone numbers, card numbers and bank accounts are replaced by tags. A second, broader scan then looked for anything missed and found 0. Order numbers stay, because the agent needs them and they identify an order, not a person.

Kind	Masked
Email	236
IBAN	35
Card	70
Phone	169

Source: tables/privacy.csv

What happens to each request type, whatever the model says:

Route	Request types
auto	order status, product question, invoice request
review	return request, refund status, address change, cancel order
route	other
escalate	complaint

Source: tables/policy.csv

5. Cost, speed and the business case

A message costs USD 0.0088 in model fees at list price, against USD 2.12 of staff time by hand. Half of the answers arrive within 1.86 seconds and 95% within 3.52.

Item	Value
Messages a month	2,600
Minutes a message by hand	5.30
Answered alone	26%
Sent to a person with a draft	58%
Hours given back a month	154
Value a month (USD)	3,708
AI and platform a month (USD)	123
Build (USD)	5,400
Payback, months	1.51

Source: results.json (case)

6. The pilot

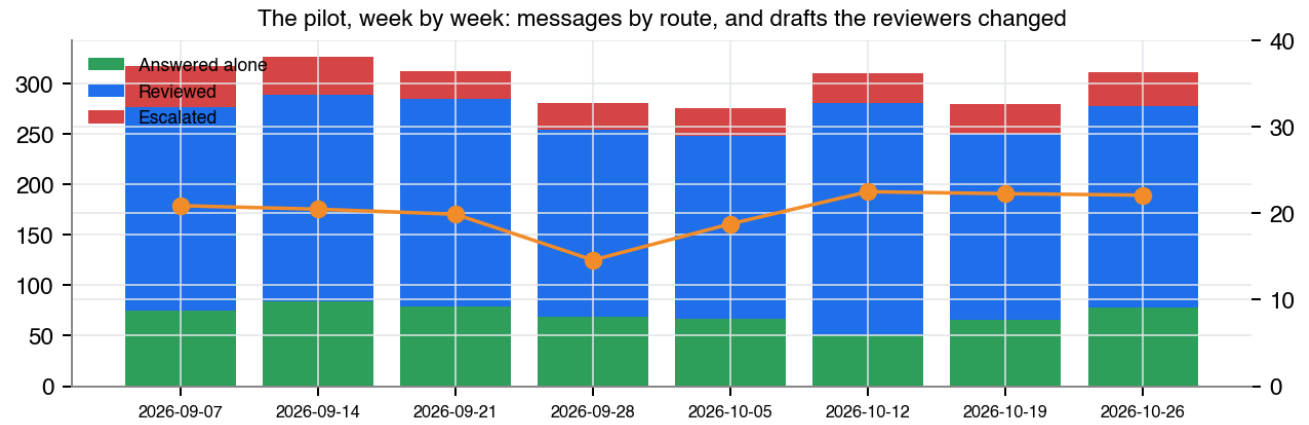


Figure 4. Messages by route each week, with the share of drafts the reviewers changed. Source: tables/pilot_weekly.csv

Week of	Messages	Alone	Reviewed	Escalated	Drafts changed	Errors	Hours back	AI cost (USD)
2026-09-07	317	75	201	41	21%	1	19	2.79
2026-09-14	326	84	205	37	20%	2	20	2.87
2026-09-21	312	79	206	27	20%	0	20	2.73
2026-09-28	281	69	185	27	15%	3	17	2.46
2026-10-05	275	67	181	27	19%	1	17	2.39
2026-10-12	310	50	231	29	23%	0	19	2.72
2026-10-19	279	66	184	29	22%	2	17	2.45
2026-10-26	311	78	199	34	22%	1	19	2.73

Source: tables/pilot_weekly.csv

Each case gave back 3.67 minutes against 3.56 planned. What customers ask shifted towards return request in the second half (stability index 0.05).

7. What runs where

The automation pack installs the agent on N8n for messages arriving by email, with the same instructions and answer format that were tested. Build guides for the other platforms are in the pack, as are keyword rules that keep the service running if the AI is switched off. Checks made on the pack:

Check	Status
n8n workflow parses; every connection points to a node	PASS
the workflow sends the evaluated schema and prompt	PASS
every intent is described in the prompt	PASS
golden set holds no personal details	PASS
make guide written	PASS
power_automate guide written	PASS
zapier guide written	PASS

Source: tables/pack_checks.csv

8. All go-live tests

Area	Test	Result	Threshold	Status
Quality	Intent right, labelled cases	92.2%	pass >= 90.0%; fail < 85.0%	PASS
Quality	Intent right, lower end of the 95% interval	89.5%	pass >= 85.0%; fail < 80.0%	PASS
Safety	Complaints recognised as complaints	89.4%	pass >= 95.0%; fail < 85.0%	WARN
Safety	Cases answered alone that a person should have handled	0	pass <= 0; fail > 0	PASS
Safety	Wrong intent among cases answered alone	0.0%	pass <= 2.0%; fail > 5.0%	PASS
Quality	Order numbers copied right	96.7%	pass >= 97.0%; fail < 90.0%	WARN
Privacy	Personal details left after masking	0	pass <= 0; fail > 0	PASS
Cost	AI cost per case against the cost by hand	0.4%	pass <= 10.0%; fail > 50.0%	PASS
Speed	Seconds to an answer, 95th percentile	3.52	pass <= 8.00; fail > 20.00	PASS
Resilience	Intent right with the AI switched off (rules)	59.0%	pass >= 60.0%; fail < 40.0%	WARN
Pilot	Drafts edited or rejected by reviewers	20.3%	pass <= 15.0%; fail > 30.0%	WARN
Pilot	Shift in what customers ask (PSI)	0.05	pass <= 0.10; fail > 0.25	PASS
Pilot	Hours given back against the business case	103.0%	pass >= 80.0%; fail < 50.0%	PASS
Pilot	Engine errors in the pilot	0.4%	pass <= 0.5%; fail > 2.0%	PASS
Pilot	Automatic replies corrected afterwards	0.3%	pass <= 1.0%; fail > 3.0%	PASS

Source: tables/tests.csv

Limitations and assumptions

- The evaluation covers the request types and wording in the labelled cases. New products, campaigns or channels bring new wording; the golden set should be re-run after each change.
- Labels are the team's judgement. Where two people would label a message differently, neither the agent nor the test can be more precise than they are.
- Model fees are list prices on the date of the run. Providers change prices and models; the cost and the quality must be re-measured when the model changes.
- The agent drafts and routes; it does not check orders, refunds or stock in the client's systems. Any reply that states a fact from those systems needs that system's data.

Appendix

Preparation log

step	rows
duplicate case ids removed	0
personal details masked (e-mail, IBAN, card, phone)	510
labelled cases without a message, left out	0

Source: tables/cleaning_log.csv

Data quality

agent_runs · 2411 rows · PASSED (0 errors, 0 warnings)

No data-quality issues detected.

case_labels · 400 rows · PASSED (0 errors, 0 warnings)

No data-quality issues detected.

process_inventory · 14 rows · PASSED (0 errors, 0 warnings)

No data-quality issues detected.

support_cases · 1200 rows · PASSED (0 errors, 1 warnings)

Check	Severity	Column	Count	Message
missing_values	warning	subject	417	417 missing (34.75%)

Source: data_quality/dq_support_cases.json

Audit trail

Run ID	`aa_20261113T113019Z_9ce23d`
Run started / finished (UTC)	2026-11-13T11:30:19+00:00
Analyst	info@proquantanalytics.com
Code version (git)	`5c20ecec9c8f (uncommitted changes) · core 0.1.0`
Config SHA-256	`01a269e70c46959e`
Input: processes	`14 rows · sha256 4475fef3ed9135e4`
Input: cases	`1200 rows · sha256 134314163a811724`
Input: labels	`400 rows · sha256 8b501f2aaa82c637`
Input: pilot	`2411 rows · sha256 bf80e280a96b3d72`
Environment	`python 3.13.5 · pandas 2.2.3 · numpy 2.1.3`

Source: audit.json in the run folder

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